

Mathematical appendix

Joint model-and-regime learning in finite-horizon market making

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Standalone appendix to the separate Bayes-adaptive research extension.

Abstract

This document gives the joint filter, exact finite-horizon recursion, model-only revelation hierarchy, information identities, matched planning interventions, and conservative interpolation and quadrature error allowances for the prespecified two-model synthetic market. Analytic inequalities concern exact expectations. Reported numerical quadrature distances and computed control tables require separate error accounting; no final economic outcome or numerical convergence claim is made here.

This is a separate extension to the core synthetic study. All results below refer to the specified synthetic model, the finite prior support

$$M \in \{1, 2\}, \quad \theta_1 = \theta_2 = 0.35, \quad (\kappa_1, \kappa_2) = (0.002, 0.10), \quad T = 30, \quad q_{\max} = 2.$$

The model label is fixed within an environment, including its independent pilot episodes. The initial regime of each episode is drawn independently with equal sign probabilities. The cold-start model probabilities are equal. Every other market, execution, accounting, and inventory convention is inherited unchanged from the task statement. None of the propositions below establishes performance in a misspecified or real market.

The mathematical quantities in this document are exact expectations and optima. Implemented tables, computed action scores, and Monte Carlo policy returns are different objects. Section 8 makes their numerical relationship explicit.

1 Decision state and chronology

Let $n = T - t$ be the number of decisions remaining. Immediately before decision t , write

$$p_{jh} = \Pr(M = j, H_t = h \mid \mathcal{I}_t), \quad w_j = \sum_h p_{jh}, \quad b_j = \Pr(H_t = +1 \mid M = j, \mathcal{I}_t).$$

The four joint masses are ordered as $(1, -1), (1, +1), (2, -1), (2, +1)$. When $w_j > 0$, $p_{j,+} = w_j b_j$ and $p_{j,-} = w_j(1 - b_j)$. At zero model weight, the corresponding conditional probability is immaterial. Calculations at that boundary must not depend on its arbitrary representation.

The sufficient decision state is (n, q, x, p) . The following order is essential:

1. Observe the current public state and select one admissible action.
2. Generate Z_t , the return, and selected passive outcomes using **the current** H_t . Book actual executions.
3. Observe the return and the outcomes of the submitted quotes. Condition the joint belief on those observations to obtain a posterior on (M, H_t) .
4. Apply each model's regime transition to obtain the pre-decision belief on (M, H_{t+1}) . Transition the public signal independently and disclose it.

Thus the observation update precedes the hidden-state prediction. A textbook POMDP convention that observes an emission from the *next* state cannot be copied into this simulator without changing the indexing.

Cash and absolute midpoint need not appear in the optimization state. If $W_t = C_t + Q_t P_t$, marked wealth satisfies the accounting identity derived in Section 3. Current wealth is an additive constant in the remaining objective; price levels and cash do not affect admissibility, observation laws, or future increments. The ledger is still required for independent execution verification. Observed cash and inventory are deterministic functions of the already observed actions and selected fills and hence supply no additional latent-state evidence.

2 Exact action-dependent joint filter

Set $\mu = 0.03$, $\sigma = 0.30$, and recover the observed innovation as $z = (r - \mu x)/\sigma$. For a submitted side $s \in \{+1, -1\}$ at depth $k_s \in \{0, 1\}$, define

$$u_s(x) = \{1 + \exp(0.3 + 0.7k_s + 0.2sx)\}^{-1}, \quad \tau_s(x) = \Phi^{-1}(u_s(x)),$$

$$\ell_h(z, f; x, a) = \prod_{s \in S(a)} \Phi\left((2f_s - 1) \frac{\tau_s(x) - h\theta sz}{\sqrt{1 - \theta^2}}\right). \quad (1)$$

Here $S(a)$ is the set of submitted passive sides and f ranges over their binary outcomes. The product is one for an empty set. A submitted non-fill is $f_s = 0$; an unsubmitted side contributes no factor. The selected sides are independent conditional on (z, h) , so their factors are multiplied **before** mixing over h . They are generally not independent after that mixture.

Both models have the same θ , so (1) depends on the model only through its current regime distribution. The full observation density with respect to dz and counting measure on f is

$$\phi(z)m(p, z, f), \quad m_j = (1 - b_j)\ell_- + b_j\ell_+, \quad m = \sum_j w_j m_j = \sum_{j,h} p_{jh} \ell_h. \quad (2)$$

The common factor $\phi(z)$ cancels in filtering, but must remain in the control expectation. The public signal transition similarly cancels from posterior odds because its known matrix K is model independent.

The posterior on the just-observed current state and the next decision belief are, respectively,

$$\tilde{p}_{jh} = \frac{p_{jh}\ell_h}{m}, \quad p'_{jh'} = \sum_h \tilde{p}_{jh} T_j(h, h'), \quad T_j = \begin{pmatrix} 1 - \kappa_j & \kappa_j \\ \kappa_j & 1 - \kappa_j \end{pmatrix}. \quad (3)$$

Equivalently,

$$w'_j = \frac{w_j m_j}{m}, \quad \tilde{b}_j = \frac{b_j \ell_+}{m_j}, \quad b'_j = \kappa_j + (1 - 2\kappa_j) \tilde{b}_j. \quad (4)$$

Denote (3) by $\Psi(p, z, f; x, a)$. For finite z , the present model has strictly positive emission probabilities, so the normalization is positive. Implementations should nevertheless use log probabilities for filtering and a defined harmless representation for underflowed zero-mass branches. An exactly excluded model stays excluded.

For abstention or a market order, $\ell_- = \ell_+ = 1$. Model weights are unchanged, and only each conditional regime prediction moves. Public returns alone contain no evidence about either latent quantity in this model.

3 Reward and exact Bellman recursion

Write $h_0 = 0.025$ for the half spread, $\Delta = 0.025$, $c_P = 0.001$, $c_T = 0.002$, and $\lambda = 0.002$, to avoid confusing the half spread with the hidden sign h . Define $d_k = h_0 + k\Delta$ and $\ell_T(q) = -(h_0 + c_T)|q|$.

For a passive action let

$$\delta q(a, f) = \sum_{s \in S(a)} s f_s, \quad c(a, f) = \sum_{s \in S(a)} f_s (d_{k_s} - c_P).$$

For a market order of sign s , set $\delta q = s$ and $c = -(h_0 + c_T)$; for abstention both are zero. In all cases,

$$W_{t+1} - W_t = c(a, f) + (q + \delta q)(\mu x + \sigma z). \quad (5)$$

The realized one-period objective increment is (5) minus λq^2 . Its conditional expectation is denoted $r(q, x, p, a)$. Let $\bar{h}(p) = \sum_{j,h} h p_{jh} = \sum_j w_j (2b_j - 1)$. Then

$$\begin{aligned} r(q, x, p, a) &= q\mu x - \lambda q^2 \\ &\quad + \sum_{s \in S(a)} [u_s(x)(d_{k_s} - c_P + s\mu x) - \sigma\theta\bar{h}(p)\phi(\tau_s(x))] \end{aligned} \quad (6)$$

for passive actions, and

$$r(q, x, p, M_s) = (q + s)\mu x - \lambda q^2 - h_0 - c_T. \quad (7)$$

To verify the covariance term in (6), conditional on $H_t = h$, the pair (Z, V_s) is standard bivariate normal with correlation $h\theta s$. Therefore $E[Z F_s \mid h, x, a] = h\theta s \int_{-\infty}^{\tau_s} v\phi(v)dv = -h\theta s\phi(\tau_s)$. Multiplication by the inventory sign s gives the last term of (6). The pre-decision inventory penalty is unchanged, including when both quotes fill.

For any continuation function F , define the common-action backup

$$(\mathcal{B}F)(q, x, p) = \max_{a \in \mathcal{A}(q)} \left\{ r(q, x, p, a) + \sum_f \int_{\mathbb{R}} \phi(z) m(p, z, f) \sum_{x'} K_{xx'} F(q + \delta q(a, f), x', \Psi(p, z, f; x, a)) dz \right\}. \quad (8)$$

The action set is safe for every possible selected-fill subset. In particular, an otherwise inadmissible bid at $q = 2$ cannot be made safe by also quoting an ask. Each branch has one future action policy based on its observed joint posterior. No model-specific maximization occurs inside the model mixture.

Proposition 1 (Bayes-adaptive optimum). The exact optimal value satisfies

$$V_0^{\text{BA}}(q, x, p) = \ell_T(q), \quad V_n^{\text{BA}} = \mathcal{B}V_{n-1}^{\text{BA}}. \quad (9)$$

Proof. Equations (1)–(4), inventory accounting, and K give the conditional law of the next sufficient state from the present sufficient state and action. Equation (5) telescopes the wealth ledger; the remaining terminal adjustment is exactly ℓ_T . Conditional expectation followed by optimization over the finite admissible action set gives (8). Backward induction proves (9). A maximizing measurable action exists at every stage; a deterministic tie rule selects one. Integrability follows from bounded inventory and finite Gaussian first moments. $\square$

For a fixed admissible policy π , its actual value V_n^π follows the same recursion with its chosen action in place of the maximum. Consequently,

$$V_n^\pi(q, x, p) \leq V_n^{\text{BA}}(q, x, p). \quad (10)$$

This is a Bayesian expectation under the displayed prior p , not a claim that the Bayes-optimal policy dominates every comparator separately in each true-model environment.

4 Exact interpretation of posterior-weighted known-model Q values

Let $V_n^j(q, x, b)$ be the exact optimum when model j is known but the current regime is hidden. Its filter still uses $\tilde{b}$ followed by the transition with κ_j . Let $Q_n^j(q, x, b, a)$ force the first action to be a , with optimal **known-model, hidden-regime** control after that action. No future shock or future regime is observed by this reference.

Consider an information relaxation that reveals M for free after the current action and its normal feedback, before the next decision. The regime remains hidden, including at the revelation time. Revealing the static model label before or after the independent hidden transition has the same effect provided the next conditional belief is (4).

Proposition 2 (one-action model revelation). Its optimal objective is

$$U_n^1(q, x, p) = \max_{a \in \mathcal{A}(q)} \sum_j w_j Q_n^j(q, x, b_j, a). \quad (11)$$

Proof. Before revelation the first action is common to both models. After normal feedback, the model has posterior probability w'_j ; if it is revealed to be j , its optimal remaining value is $V_{n-1}^j(q + \delta q, x', b'_j)$. The branchwise identity $mw'_j = w_j m_j$ gives

$$\begin{aligned} r(q, x, p, a) + E \left[\sum_j w'_j V_{n-1}^j(q + \delta q, X', b'_j) \right] \\ = \sum_j w_j Q_n^j(q, x, b_j, a). \end{aligned} \quad (12)$$

Maximizing the common first action proves (11). $\square$

The information available to the relaxed controller contains that of the real controller. It can ignore its additional label and imitate any admissible real policy. Combining this observation with (10) gives

$$\boxed{V_n^\pi \leq V_n^{\text{BA}} \leq U_n^1.} \quad (13)$$

The executable weighted-Q policy selects the maximizing action in (11), updates its real posterior, and repeats at the next decision. **Its realized value is $V^{\pi^{\text{wQ}}}$, not U^1 :** the promised model revelation never actually arrives. Weighted-Q can learn the model through its online filter and can value regime feedback through each Q^j . Its planning simplification is that future *model uncertainty* is removed after the current action.

Equation (13) requires exact optimal known-model Q functions. If those functions are heuristics or uncertified numerical approximations, the raw computed maximum is not automatically an upper bound. A regime-observing oracle also defines a different, more informative comparator than (11).

5 A hierarchy of delayed-revelation bounds

For an integer $d \geq 0$, reveal the model after exactly d further decisions, before any later decision. Let U_n^d be the optimum in that relaxed problem. Revelation at $d = 0$ precedes the current decision, so

$$U_n^0(q, x, p) = \sum_j w_j V_n^j(q, x, b_j), \quad U_0^d(q, x, p) = \ell_T(q), \quad (14)$$

and for $d \geq 1, n \geq 1$,

$$U_n^d = \mathcal{B}U_{n-1}^{d-1}. \quad (15)$$

The delay decreases after each unrevealed decision. Recursing on U_{n-1}^d with the same delay would define a different problem.

Proposition 3 (nested information sets). For all states and n ,

$$U_n^0 \geq U_n^1 \geq U_n^2 \geq \dots \geq U_n^n = U_n^{n+1} = \dots = V_n^{\text{BA}}. \quad (16)$$

Proof. A controller told the model after d actions can ignore it until after action $d + 1$ and implement any policy from the later-revelation problem. The physical dynamics and permissible actions are the same, proving each inequality. If $d \geq n$, no remaining decision can use the label; terminal liquidation is fixed and model independent. Such a label changes no payoff, proving equality to (9). Equivalently, induction in (15) reaches the common terminal condition before any useful revelation. $\square$

This proves a hierarchy of *objectives*. It does not prove a monotone ranking of the real returns of the policies greedy with respect to those objectives. Their remaining optimism and resulting action errors can differ. Delays one, two, and three at horizon 30 are limited-lookahead information relaxations; they are not full Bayes-adaptive optima at horizon 30. At horizon two, delay two must agree with the exact Bayes-adaptive recursion; at horizon three, delay three must agree.

Useful exact limiting cases are a singleton model posterior; two identical models; and $\theta = 0$, where the hidden process has no execution or reward effect. In each case the economically relevant model uncertainty disappears. For a singleton posterior every delay agrees with the corresponding known-model hidden-regime optimum.

For a genuine feasible-policy lower value $L \leq V^{\text{BA}}$ and a certified upper value $U \geq V^{\text{BA}}$, $U - L$ bounds the remaining optimality gap. A simulation estimate of V^π and an uncertified interpolated U^d do not, by themselves, supply such a certificate. Also, (16) concerns the prior-weighted population. A realized episode or a true-model-specific mean can exceed a prior-averaged relaxation value without contradicting the theorem.

6 What an observation teaches about the model and the regime

6.1 The two information quantities share an observation channel

Condition throughout on the pre-decision information and selected action. With common θ , the emission depends on H_t , not directly on M :

$$M \longrightarrow H_t \longrightarrow O_t. \quad (17)$$

This conditional Markov chain implies

$$I(M, H_t; O_t) = I(H_t; O_t) = I(M; O_t) + I(H_t; O_t \mid M). \quad (18)$$

The two summands give a valid information decomposition, not an additive decomposition of economic value. Their numerical forms include

$$\mathcal{I}_M(p, a) = \sum_f \int \phi(z) m \sum_j w'_j \log \frac{w'_j}{w_j} dz, \quad (19)$$

$$\mathcal{I}_{H|M}(p, a) = \sum_j w_j \sum_{h \in \{-1, +1\}} b_{jh} \sum_f \int \phi(z) \ell_h \log \frac{\ell_h}{m_j} dz, \quad (b_{j,-}, b_{j,+}) = (1 - b_j, b_j). \quad (20)$$

Zero-weight terms are interpreted by continuity. These are nonnegative expected information gains. A particular realized posterior entropy can increase. Entropy computed after predicting H_{t+1} additionally includes uncertainty introduced by switching and is not the information acquired about H_t . Conditional regime information can be lost through that transition; the model posterior is unchanged by prediction because the model is static.

The sign-free marginal fill law alone cannot identify either H_t or κ here. This also holds for the two selected side outcomes observed without their return: simultaneous sign reversal is absorbed by the symmetric Gaussian innovation, and their joint law depends on θ^2 , not the regime sign. Returns alone are likewise uninformative. Their **joint** observation is informative. These statements condition on the known action and public signal, rather than treating a policy's selected actions as an independent evidence source.

6.2 Cold start has exactly zero first-observation model information

At cold start $b_1 = b_2 = 1/2$. For every action and every observation, $m_1 = m_2$, and hence $w'_j = w_j$ and $\mathcal{I}_M(p, a) = 0$. More generally this holds whenever the two conditional current-regime beliefs coincide. The first observation may nevertheless update their common current-regime posterior $\tilde{b}$. Subsequent prediction gives

$$b'_1 = 0.002 + 0.996\tilde{b}, \quad b'_2 = 0.10 + 0.80\tilde{b}. \quad (21)$$

Unless $\tilde{b} = 1/2$, those probabilities differ. A later informative observation can then distinguish persistence models. Initial information acquisition can prepare a later model-learning opportunity even though its own immediate model information gain is zero. A one-period pilot episode therefore supplies no information about κ in this two-model family.

The absence of first-step model information is not a reason for the optimal initial action to match weighted-Q. The policies make different assumptions about information and uncertainty at future decisions.

6.3 Complete regime evidence generally cannot be kept while model evidence is erased

For positive likelihoods, let $R = \ell_+/\ell_-$. The conditional regime posterior odds multiply by R , while the model Bayes factor is

$$\frac{m_1}{m_2} = \frac{1 - b_1 + b_1 R}{1 - b_2 + b_2 R}. \quad (22)$$

When $b_1 \neq b_2$, the right side is strictly monotone in R : its derivative is $(b_1 - b_2)/(1 - b_2 + b_2 R)^2$. Thus preserving the full likelihood-ratio evidence needed by both regime filters also preserves the model Bayes factor. Except at special states or uninformative observations, there is no coherent observation deletion that removes model evidence while retaining all that regime evidence. This is a structural limitation of the proposed attribution experiment, not a numerical defect.

7 Matched suppression of model-weight updates in planning

A useful controlled computational intervention is still available. Starting from the same actual belief (w, b_1, b_2) , compute the true predictive outcome probability and the true conditional regime updates in (4). Replace only the next model weights by the current ones when reading a continuation value:

$$\Psi_{\text{FM}}(p, z, f) = (w_j(1 - b'_j), w_j b'_j)_{j=1,2}. \quad (23)$$

For a fixed common continuation F , define

$$Q_F^{\text{BA}}(a) = r + E_p[F(q', X', \Psi)], \quad Q_F^{\text{FM}}(a) = r + E_p[F(q', X', \Psi_{\text{FM}})]. \quad (24)$$

The two expectations use identical true current outcome probabilities, immediate rewards, inventory transitions, legal actions, and continuation function. Only the model-weight update at the next planning boundary changes. Using $F = V_{n-1}^{\text{BA}}$ makes (24) a direct action-level diagnostic without changing the downstream continuation function at the same time.

For the executable recursive ablation, let $W_0 = \ell_T$ and apply (23) at every hypothetical step to construct W_n . Its policy reads these scores from the **true current posterior**, and its online filter uses (3), just as the other feasible policies do. Thus it suppresses future model-weight updates in planning, not actual permitted feedback at deployment. Distinguish this recursive policy comparison from the single-backup experiment in (24).

Important limits are part of its definition:

- Ψ_{FM} is an artificial belief intervention, not the Bayes posterior for the physical observation experiment. It defines a planning surrogate; no upper- or lower-bound claim attaches to W_n or to the score difference in (24).
- The conditional regime filters b'_j are retained, but the marginal regime probability changes from $\sum_j w'_j b'_j$ to $\sum_j w_j b'_j$. Subsequent forecasts can therefore change along with model learning. By (22), a completely clean separation generally cannot be realized while preserving all regime evidence.
- Neither each realized difference in (24) nor its expectation must be nonnegative. Jensen's theorem for coherent posterior conditioning does not apply to (23).
- At a state with $b_1 = b_2$, (23) agrees with the true update for every current observation. The one-backup diagnostic must then be exactly zero for any fixed F . Recursive BA and frozen-model policies can still disagree there because their future continuation functions differ.

For comparison, a prediction-only intervention freezes both the observation update of w and that of each b_j , reading continuation at $b'_j = \kappa_j + (1 - 2\kappa_j)b_j$. It still integrates the actual stochastic inventory outcomes. It changes more than the model-weight intervention and cannot isolate its contribution.

An action-level finding should retain the complete state, admissibility mask, all action scores under (24), immediate rewards, numerical refinements, and model/regime information diagnostics. An action difference stable at relevant numerical resolutions supports sensitivity to the declared planning update. Fresh paired objective comparisons are needed to establish economic value. Lower entropy, a selected action disagreement, or a negative immediate reward alone establishes neither model-learning benefit nor profitable exploration. If the specified contrast is negligible, adverse, or numerically unstable, that is the finding; it is not a reason to relabel regime information as model information or to select a new preferred hypothesis after evaluation.

8 Numerical error: what can be proved and what remains empirical

8.1 Convexity and the product coordinates

For a fixed (n, q, x) and a fixed nonanticipative policy tree, the expected remaining payoff is linear in the initial four-state mass vector p . Taking the supremum over the same class of admissible policy trees proves convexity of V_n^{BA} in p . The same argument holds for U_n^d , where the policy tree also has its prescribed future revelation. Continuous observations mean that this need not be a maximum of finitely many linear functions; a finite latent state space alone does not justify a finite piecewise-linear representation.

On a rectangular grid in (w, b_1, b_2) , with $w = w_1$, the mapping

$$p(w, b_1, b_2) = (w(1 - b_1), wb_1, (1 - w)(1 - b_2), (1 - w)b_2) \quad (25)$$

is multiaffine. The nonnegative tensor interpolation weights α_v on the eight corners satisfy $p(w, b_1, b_2) = \sum_v \alpha_v p_v$. Therefore, for exact nodal values of any convex joint-belief value V ,

$$V(p) \leq \sum_v \alpha_v V(p_v). \quad (26)$$

This fact concerns joint-mass barycenters; convexity in the three conditional coordinates is not assumed. With exact integration, exact maximization, and nonnegative interpolation, an induction from the terminal payoff makes the gridded BA calculation an upper approximation. Positive quadrature weights preserve monotonicity but **do not give a sign to Gaussian quadrature error**. Thus (26) alone does not certify the implemented Gauss–Hermite tables.

8.2 A global payoff and interpolation envelope

A conservative exact bound for any conditional expected stage reward is $|r(q, x, \delta_{jh}, a)| \leq r_* = 0.231$. To check it, $|q\mu x| \leq 0.06$, $\lambda q^2 \leq 0.008$, $u_s < 1/2$, $|d_k - c_P + s\mu x| \leq 0.079$, and $\sigma\theta\phi(\tau_s) < 0.042$. There are at most two submitted sides. Market actions have a smaller bound. Terminal liquidation lies in $[-0.054, 0]$. Consequently every policy tree’s expected payoff, conditional on any initial hidden state, lies in

$$[-0.231n - 0.054, 0.231n].$$

Let $D_n = 0.231n + 0.027$, half this common payoff envelope’s width. Subtracting its midpoint from each linear policy coefficient proves

$$|V_n(p) - V_n(\bar{p})| \leq D_n \|p - \bar{p}\|_1 \quad (27)$$

for exact BA and fixed-delay revelation values at fixed (q, x) . This uses a bound on **conditional expected** stage rewards, not an incorrect bounded-support assertion about Gaussian realized PnL.

If the containing cell has local widths $\Delta_w, \Delta_{b_1}, \Delta_{b_2}$, the tensor corner distribution gives

$$\sum_v \alpha_v \|p_v - p\|_1 \leq \Delta_w + w\Delta_{b_1} + (1-w)\Delta_{b_2} \leq \Delta_w + \max_j \Delta_{b_j}. \quad (28)$$

For example, split the difference by first changing w and then b_1, b_2 . The respective ℓ_1 changes are $2|\delta w|$ and $2[w|\delta b_1| + (1-w)|\delta b_2|]$. Linear endpoint sampling has expected absolute displacement at most half its interval width, proving (28). Equations (26)–(28) bound interpolation error by $D_n(\Delta_w + \max_j \Delta_{b_j})$.

8.3 A global Gaussian quadrature envelope without dividing by a tiny posterior mass

Set

$$c_\theta = \frac{\theta}{\sqrt{1 - \theta^2} \sqrt{2\pi}} < 0.15.$$

Each conditional Bernoulli fill probability is c_θ -Lipschitz in z . For $s_a \leq 2$ submitted sides, the ℓ_1 distance between the two product Bernoulli outcome distributions at z and $\bar{z}$ is at most $2s_a c_\theta |z - \bar{z}|$.

For each fill branch, use its **unnormalized** next-state vector $v_f(z)_{jh'} = \sum_h p_{jh} \ell_h(z, f) T_j(h, h')$. It has mass m . The homogeneous value perspective $mV(q', x', v_f/m)$ is the supremum of linear policy payoffs in v_f . After subtracting the common payoff-envelope midpoint times the mass, it is D_n -Lipschitz in v_f . Hidden-state prediction contracts ℓ_1 distance. Summing over fills and averaging over K therefore shows that the exact continuation integrand

$$g(z) = \sum_f m(p, z, f) \sum_{x'} K_{xx'} V_n(q', x', \Psi(p, z, f))$$

is $2s_a c_\theta D_n$ -Lipschitz. The midpoint terms cancel because $\sum_f m(p, z, f) = 1$ at every z . This proof avoids any lower bound on individual posterior denominators.

Let $\nu = \sum_i \omega_i \delta_{z_i}$ be a normalized positive numerical quadrature law and let $\mathcal{W}_1 = W_1(\mathcal{N}(0, 1), \nu)$. Coupling the two scalar variables gives the deterministic inequality

$$\left| E[g(Z)] - \sum_i \omega_i g(z_i) \right| \leq 2s_a c_\theta D_n \mathcal{W}_1 \leq 4c_\theta D_n \mathcal{W}_1. \quad (29)$$

This applies even though continuation has decision kinks. It requires no unjustified high-order derivative bound for Gauss–Hermite quadrature.

The transport distance has a one-dimensional exact expression. Sort the nodes, put $c_i = \sum_{k \leq i} \omega_k$, and $a_i = \Phi^{-1}(c_i)$, with $a_0 = -\infty$ and $a_N = +\infty$. Then

$$\mathcal{W}_1 = \sum_{i=1}^N \int_{a_{i-1}}^{a_i} |z - z_i| \phi(z) dz. \quad (30)$$

Each term can be evaluated by splitting at z_i and using $\int z \phi(z) dz = -\phi(z)$. Floating evaluation for ideal normalized Gauss–Hermite rules gives approximately 0.2020035, 0.1567188, 0.1224879, and 0.1004662 for 15, 25, 41, and 61 nodes. These displayed evaluations are not outward-rounded interval certificates for the stored floating-point rule.

8.4 Propagation, one-sided control, and the size of the allowance

Bellman maximization and expectation are nonexpansive in the uniform norm. With one grid, quadrature rule, exact analytic rewards, and exact arithmetic, let δ_G be the maximum over all grid cells of $\Delta_w + \max_j \Delta_{b_j}$. This includes nonuniform meshes. Starting from the exact terminal value, the error at computed grid nodes obeys

$$\max_{q,x,p \text{ on grid}} |\hat{V}_n(q, x, p) - V_n^{\text{BA}}(q, x, p)| \leq \sum_{k=0}^{n-1} D_k (\delta_G + 4c_\theta \mathcal{W}_1). \quad (31)$$

To see this, interpolate the exact preceding-horizon nodal values, apply (28), and add (29). The previous uniform nodal error passes through the positive normalized operator without amplification. The maximum over actions cannot increase that error. Off-grid value queries can require an additional $D_n \delta_G$ for the last interpolation; this term must not be omitted from a claimed two-sided guarantee.

There is a useful one-sided improvement: interpolation of the exact value never lowers it. At every backup, compare the computed continuation to the exact value using (26) before applying (29). Induction proves, including for an interpolated final value,

$$V_n^{\text{BA}}(q, x, p) \leq \hat{V}_n(q, x, p) + \sum_{k=0}^{n-1} 4c_\theta D_k \mathcal{W}_1. \quad (32)$$

The bound has no interpolation penalty on this side. Analogous arguments apply to delayed-revelation recursions if the terminal known-model values carry their own applicable error allowance. They must not be assumed exact merely because they came from the frozen known-model solver.

These formulas are a route to a conservative certificate, not evidence that a tight certificate has been implemented. Validating floating normalization, special-function evaluations, all arithmetic, and tie tolerance adds its own per-stage allowance. A chosen action whose score is within the 10^{-12} tie tolerance of the maximum adds at most that tolerance per exact backup; floating arithmetic requires a separately justified allowance.

The magnitude matters. At $n = 30$, $\sum_{k=0}^{29} D_k = 101.295$. Using the displayed 61-node transport distance makes the allowance in (32) about **6.07 objective units**, before floating-point certification. For illustration only, a uniform 33-by-65-by-65 grid has $\delta_G = 0.046875$, adding about 4.75 units to the two-sided nodal envelope (31). Any other mesh uses its own maximum cell-width sum. These safe-envelope calculations are far too loose to certify a 0.002 economic gap. Their existence does not justify calling a much smaller observed refinement difference a rigorous error bound.

For an action choice, a uniform ε bound on every action score would certify a unique leading action when its estimated lead exceeds 2ε . It would bound the chosen action’s true one-step loss by 2ε , and such losses could be accumulated over a finite horizon. No such small uniform ε follows from a few checked states.

The practical validation must therefore report grid and quadrature refinement, short-horizon exact-recursion checks, singleton-model agreement, reached-state action stability, and independently integrated Bellman residuals as **numerical evidence**. Pointwise residuals do not control unsampled states. Monte Carlo intervals address sampling variability of an executable policy under the evaluation population; they do not cover solver bias or supply an

optimality gap. Plots should identify $\widehat{U}^d$ as numerical relaxation estimates, separately from achieved-policy means and their sampling intervals.

9 Pilot episodes, reset semantics, and validation identities

For independent pilot episodes $D_1, \dots, D_E$ from the same fixed model,

$$w_j(D_{1:E}) \propto w_j^{\text{prior}} \prod_{e=1}^E L_j(D_e), \quad L_j(D_e) = \sum_{h_0, \dots, h_{T-1}} \frac{1}{2} \left[\prod_{t=0}^{T-1} \ell_{h_t}(z_t, f_t) \right] \left[\prod_{t=0}^{T-2} T_j(h_t, h_{t+1}) \right]. \quad (33)$$

Common return and signal densities are omitted only because they cancel in model odds. The pilot action probabilities also cancel: the behavior policy is known and chooses uniformly among publicly admissible actions. Conditioning on the realized actions does not license access to unselected fills.

Each episode starts the conditional regime filter at $b_j = 1/2$. Accumulate its model log evidence, retain model weights across pilot episodes, and reset regime beliefs at the next independent episode. Carrying the final regime posterior into the next independent pilot fabricates persistence; resetting model weights discards valid shared-model evidence.

Every fresh evaluation episode starts at $q = 0, x = 0, p_{jh} = w_j(D_{1:E})/2$. If the declared estimand is performance after a fixed pilot budget, evaluation episodes independently reuse that same pilot posterior. They do not gradually increase each other's training budget. All feasible policies receive the identical pilot within a replicate. Model labels and exogenous tapes belong to the evaluator, not the policy interface.

Useful independent checks of the derivation are:

- Joint prediction equals (3), and its model marginal equals (4).
- All selected-fill branches integrate to one; market/abstention updates change no model weights; the unused conditional belief at zero weight has no effect.
- A one-period evaluation uses $\ell_T(q')$, not a second liquidation; post-terminal hidden prediction cannot change its value. Every belief-update suppression has identical one-period scores when it shares the actual inventory outcomes, because the terminal continuation is independent of belief.
- Singleton-model BA values agree with the known-model hidden-regime recursion. Identical models collapse in the same way.
- Delay one equals weighted exact known-model Q action by action; delay n agrees with BA at short horizon n .
- Model information and the single-backup frozen-weight score difference are exactly zero when $b_1 = b_2$, including cold start.
- The online policy value, the numerical Bellman table, and a model-revelation objective are never substituted for one another in reports.

These checks belong to the predeclared E2E validation artifact. They are not a request to add post-implementation unit tests to the frozen core.

10 Attribution to established work

The Bayes-adaptive idea—include uncertainty about the environment model in the hidden state and plan jointly for model learning, state identification, and reward—is established. Ross, Chaib-draa, and Pineau formulate it for uncertain transition and observation probabilities using count-augmented states. The present two-point prior is a finite, simpler instance of the same decision principle; it does not require their Dirichlet construction. Ross, Chaib-draa and Pineau (2007), *Bayes-Adaptive POMDPs*, Sections 2–3.

The use of extra information to upper-bound partially observed control and the distinction between an approximate value and the control policy extracted from it are also established. Hauskrecht analyzes QMDP, bound hierarchies, and convex interpolation. Our weighted-Q interpretation is related to QMDP but reveals **only the model**, while its regime remains hidden. Equations (11)–(16) explicitly derive the appropriate information relaxation for this task instead of importing the fully state-observing QMDP formula. Hauskrecht (2000), *Value-Function Approximations for Partially Observable Markov Decision Processes*, Sections 3.2, 4.1–4.2, 4.5.

Smallwood and Sondik establish finite-horizon belief-space control and the convex piecewise-linear structure in their finite formulation. We use the policy-as-linear-payoff argument to establish convexity directly for this task’s continuous return observation, without claiming finitely many value pieces. The filter, accounting specialization, model-only revelation proof, delayed hierarchy, common-emission attribution limitation, and error envelopes above are worked derivations for this particular extension; no general-method novelty is claimed. Smallwood and Sondik (1973), *The Optimal Control of Partially Observable Markov Processes over a Finite Horizon*.